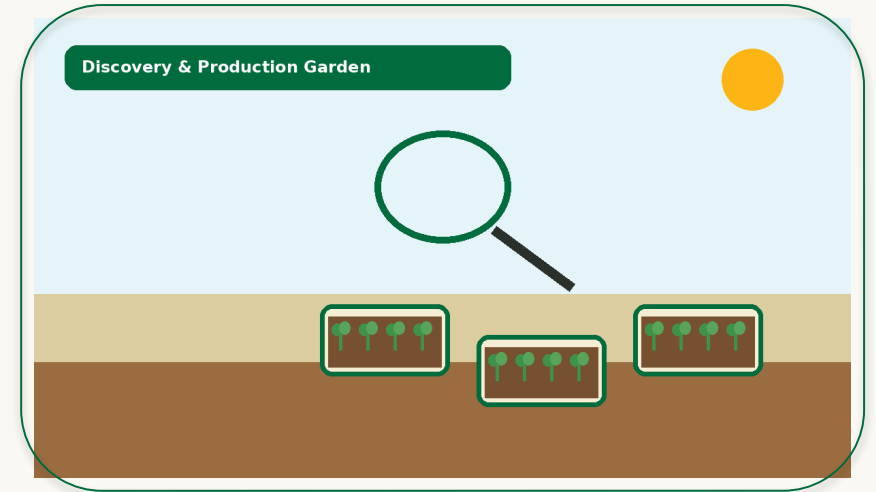


Discovery & Production Garden

From Investigation to Harvest

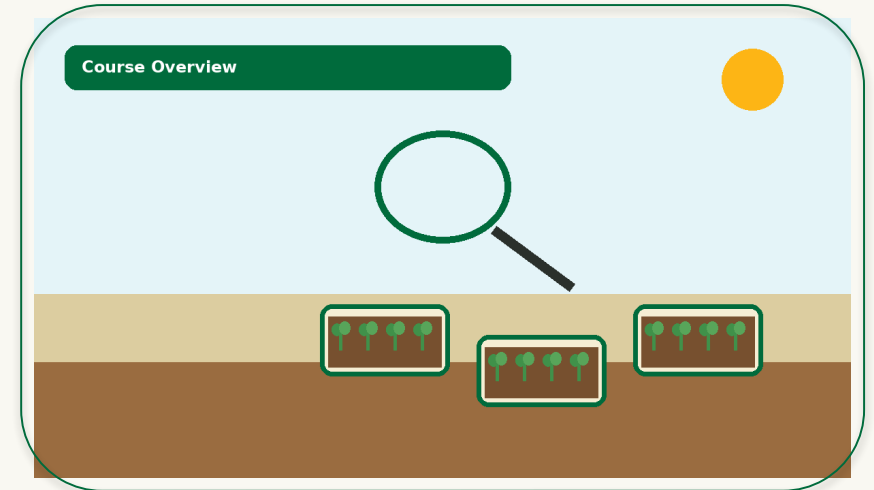
- LIFEWS AgriShine™ GrowMeal™ Primary 1-3 Teacher Training
- Audience: Primary 1-3 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised practical orientation
- Recommended duration: 3 hours online + 60-90 minutes garden practical



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden's educational and production purpose; teach the ten course topics accurately; connect practical gardening to Primary 1-3 curriculum; supervise age-appropriate garden work; maintain simple records; assess learning through practical and classroom evidence.
- Core materials: teacher guide, crop/word/number cards as relevant, child-safe hand tools, watering equipment, labels, clipboards, rulers/measuring tools where relevant, observation/record sheets, clean harvest containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe garden lesson, and submit a short lesson/assessment plan.



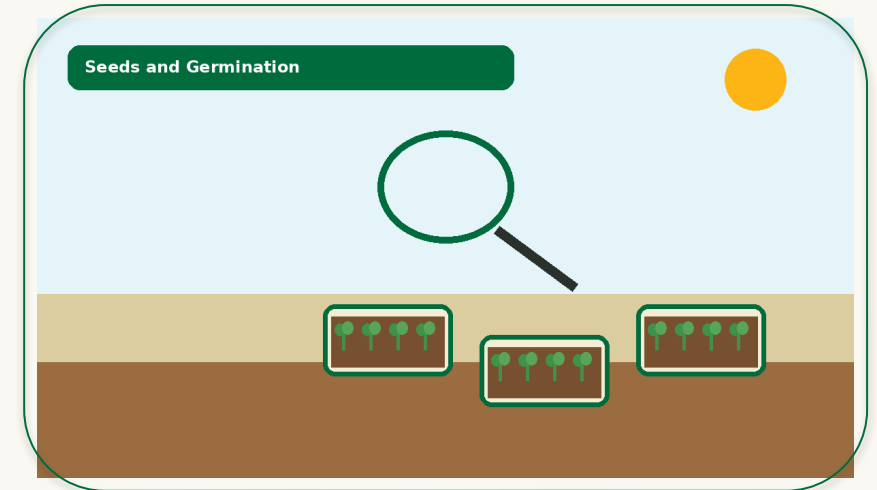
TEACHER PRACTICE

- Training sequence: EXPLAIN → DEMONSTRATE → PRACTISE → RECORD → REFLECT.

Seeds and Germination

Teacher knowledge and practical application

- Compare seeds of selected food crops and observe germination using safe demonstrations.
- Introduce the sequence seed → germination → seedling.
- Discuss suitable water, air, temperature and growing conditions in simple terms.
- Avoid repeatedly digging up planted seeds.
- Record emergence dates where appropriate.



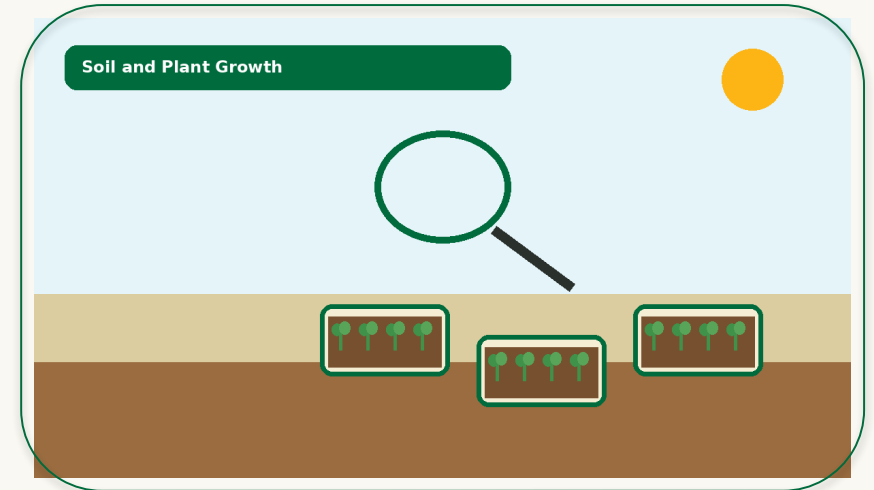
TEACHER PRACTICE

- Prepare a seed-to-seedling observation.

Soil and Plant Growth

Teacher knowledge and practical application

- Observe soil as the root environment and compare safe samples by texture, moisture or organic matter appearance.
- Explain that soil conditions influence plant growth.
- Avoid declaring one sample 'good' or 'bad' from appearance alone.
- Connect roots, water and soil through real plants.
- Use simple records.



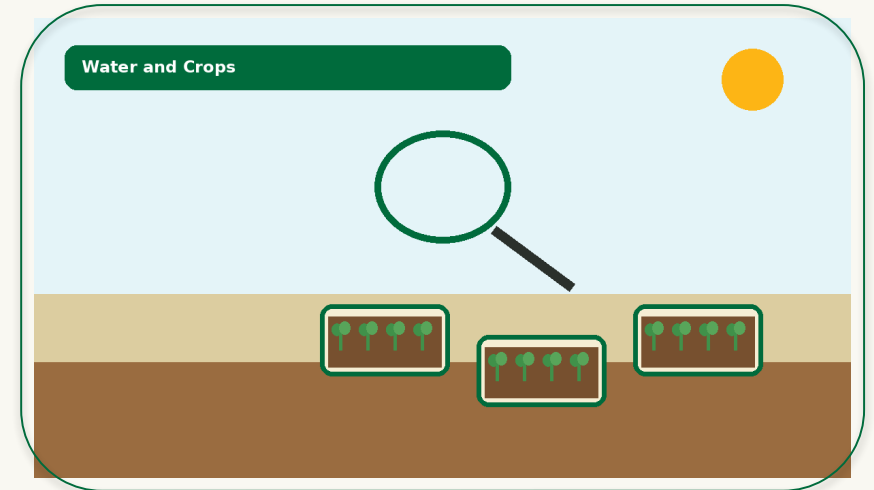
TEACHER PRACTICE

- Prepare two safe soil observations.

Water and Crops

Teacher knowledge and practical application

- Show how crops receive water and why both shortage and excess can be harmful.
- Observe moisture before and after watering.
- Trace the path from water source to root zone.
- Use simple measurement where available.
- Connect water care to production.



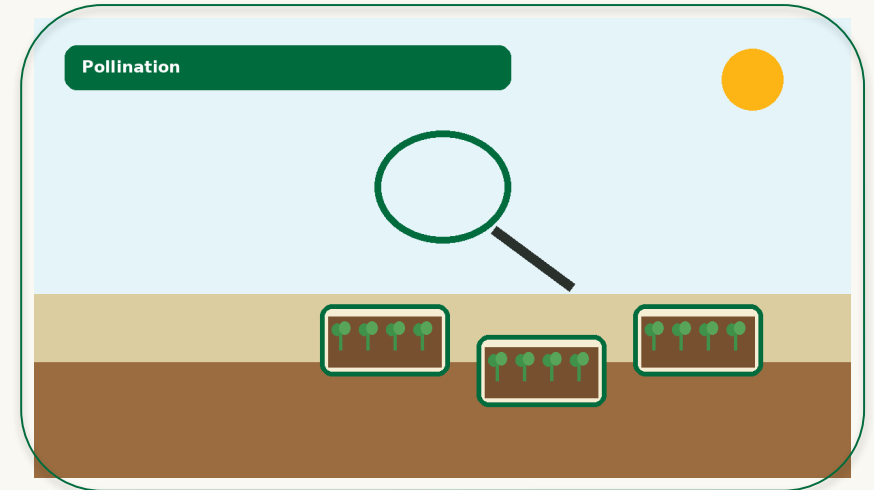
TEACHER PRACTICE

- Prepare a water-path activity.

Pollination

Teacher knowledge and practical application

- Observe flowers and garden visitors on selected food crops.
- Explain the role of pollination for many fruit- and seed-producing crops.
- Use models when pollinators are absent.
- Maintain safe distance from insects.
- Connect flowering observations to later fruit/seed development.



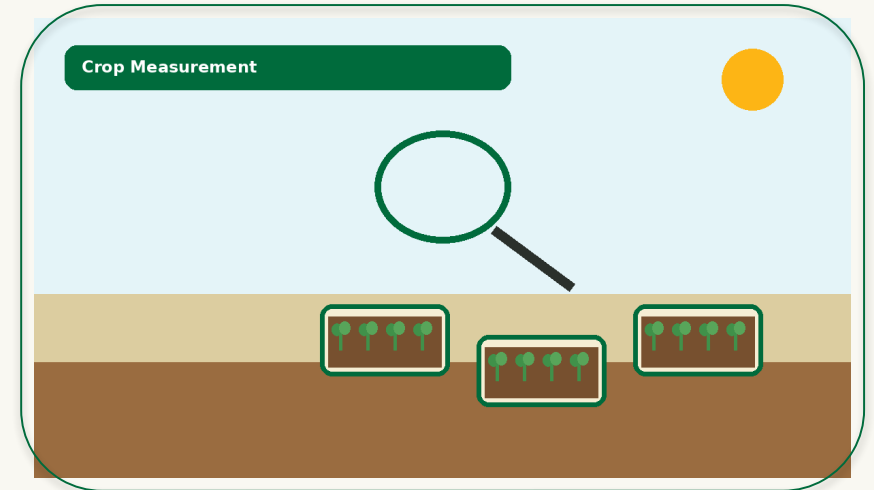
TEACHER PRACTICE

- Prepare a flower-to-fruit observation sequence.

Crop Measurement

Teacher knowledge and practical application

- Measure height, leaf number, plant count or other age-appropriate indicators.
- Use consistent methods and units.
- Record date and crop.
- Repeat measurements to show change.
- Avoid measuring in ways that damage plants.



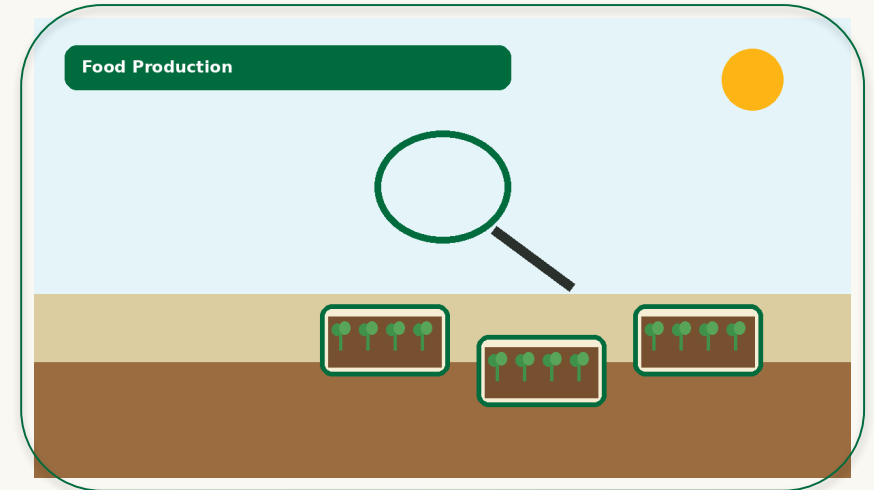
TEACHER PRACTICE

- Create a three-measurement crop record.

Food Production

Teacher knowledge and practical application

- Connect planning, planting, care, monitoring and harvest as parts of production.
- Discuss resources required: soil/growing medium, water, labor, seed and space.
- Assign pupils age-appropriate production roles.
- Use records to support decisions.
- Emphasize care and responsibility rather than maximum yield alone.



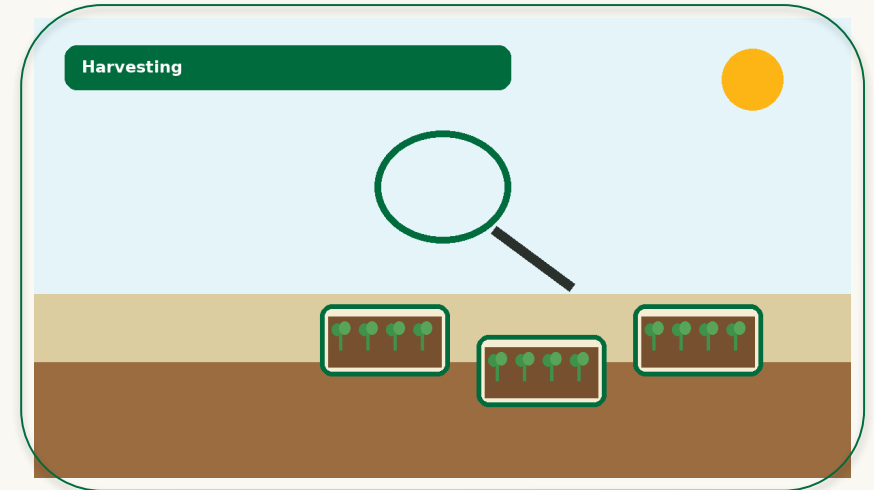
TEACHER PRACTICE

- Map the class crop-production cycle.

Harvesting

Teacher knowledge and practical application

- Identify readiness, demonstrate safe harvest and use clean containers.
- Count or weigh produce where appropriate.
- Record what was harvested.
- Discuss what happens next: eating, cooking, sharing, storage or school use according to program rules.
- Follow food safety.



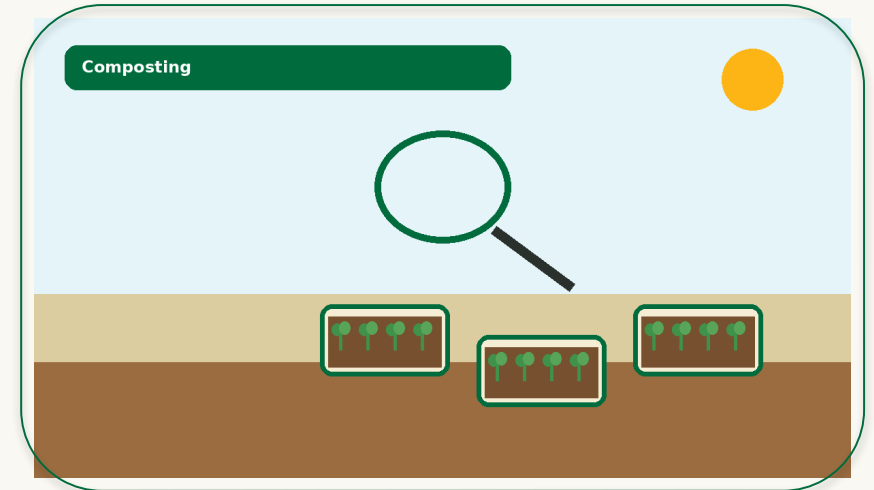
TEACHER PRACTICE

- Prepare a harvest-to-use flow activity.

Composting

Teacher knowledge and practical application

- Introduce composting as managed decomposition of suitable organic materials.
- Use only established school compost systems.
- Distinguish acceptable garden/food materials from prohibited waste.
- Children do not handle fresh manure or unsafe decomposing material.
- Use mature compost in production according to garden guidance.



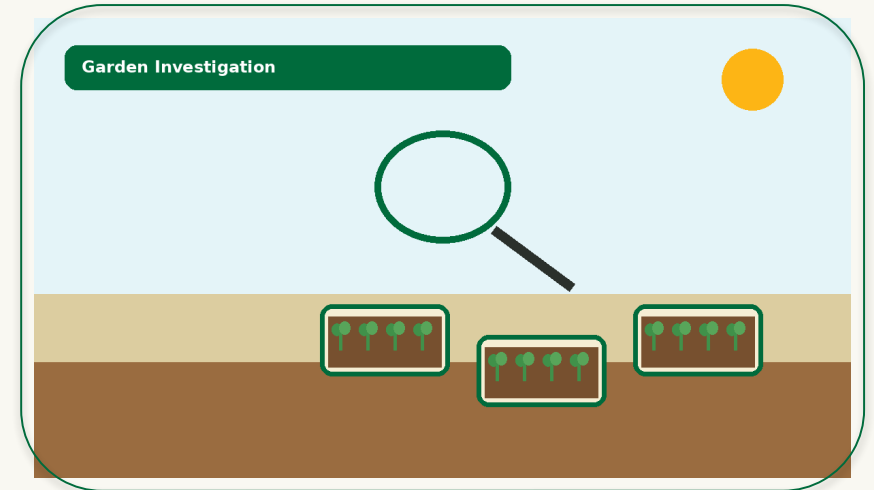
TEACHER PRACTICE

- Prepare a safe compost-system orientation.

Garden Investigation

Teacher knowledge and practical application

- Begin with a simple question that can be answered through observation or measurement.
- Examples: Which row has more emerged seedlings? Which bed stays moist longer after watering?
- Keep comparisons fair where possible.
- Record evidence before discussing explanations.
- Avoid presenting one small observation as universal proof.



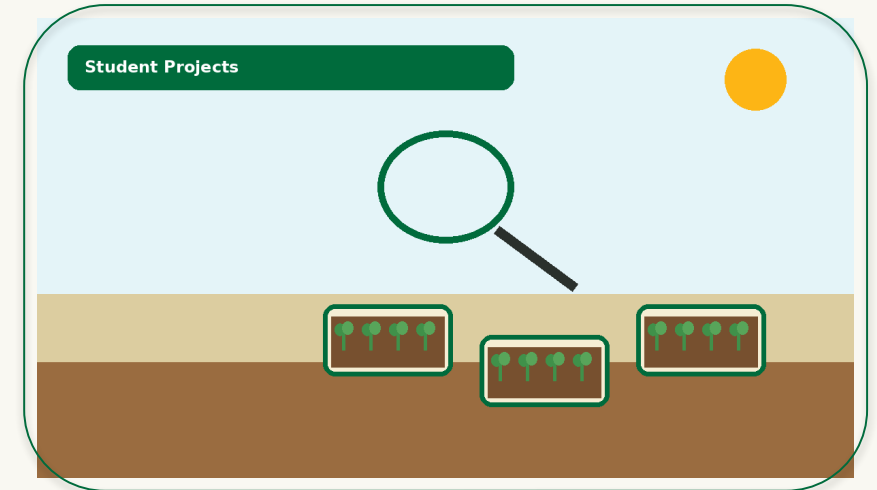
TEACHER PRACTICE

- Prepare one question-measure-record investigation.

Student Projects

Teacher knowledge and practical application

- Use short projects combining a question, practical task, observation record and presentation.
- Examples include seed germination diary, crop growth chart, water-use observation or harvest record.
- Define roles and safety requirements.
- Assess both process and learning product.
- Keep projects achievable within the school term.



TEACHER PRACTICE

- Completion task: design a 2-4 week Discovery & Production project.